Reducibility 08

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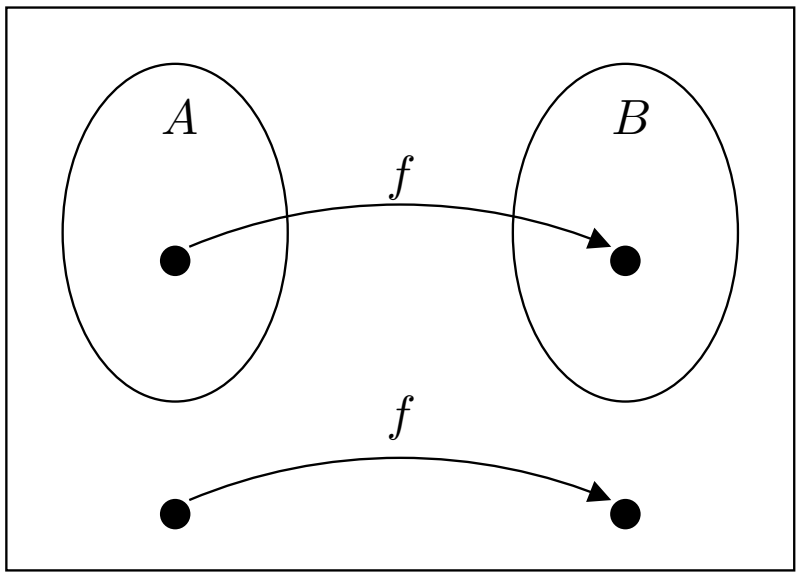
Mapping Reducibility

Definition 5.20

Language A is **mapping reducible** to language B, written $A \leq_m B$, if there is a computable function $f : \Sigma^* \to \Sigma^*$, where for every w,

$$w \in A \leftrightarrow f(w) \in B.$$

The function f is called the **reduction** from A to B.



Function f reducing A to B

Reductions

- Crucial Features in a Reduction:
  1. For every instance w of the first problem that we start with, we must be able to obtain the instance $f(w)$ of the second problem **algorithmically**
     - In our discussion, this **reduction** $f : \Sigma^* \to \Sigma^*$ will be a TM that produces an output string
  2. The answer to the second question for the instance $f(w)$ must be the same as the answer to the original question for w

     $$w \in A \leftrightarrow f(w) \in B.$$

     In other words, for every w,
     - If $f(w) \in B$, w must be in A
     - If $f(w) \notin B$, w must not be in A
  3. The **reduction** does not have to be one-to-one or onto as long as it satisfies the above property

Mapping Reducibility

Theorem 5.22

If $A \leq_m B$ and B is decidable, then A is decidable.

Proof:

- Assume that $A \leq_m B$ and B is decidable
- Since $A \leq_m B$, there exists a reduction in the form of a TM F satisfying

$$w \in A \leftrightarrow F(w) \in B$$

- Since B is decidable, there exists a TM M_B that decides B
- We just need to use the same method as in Chapter 4 to show that A is decidable

Mapping Reducibility

- Construct a decider N (to decide A) as follows:

  $N =$ "On input w:

  1. Run TM F (a reduction from A to B) on w to produce $F(w)$.
  2. Run M_B (a decider for B) on $F(w)$
  3. If M_B accepts $F(w)$, *accept*. If M_B rejects $F(w)$, *reject*."

- Prove that if $w \in A$, N accepts w:

  Assume that $w \in A$. Since $w \in A$, from the property of the reduction F, $F(w) \in B$. Since $F(w) \in B$ and M_B is a decider for B, by running M_B on input $F(w)$, M_B will accept $F(w)$. Since M_B accepts $F(w)$, N accepts w.

- Prove that if $w \notin A$, N rejects w:

  Assume that $w \notin A$. Since $w \notin A$, from the property of the reduction F, $F(w) \notin B$. Since $F(w) \notin B$ and M_B is a decider for B, by running M_B on input $F(w)$, M_B will reject $F(w)$. Since M_B rejects $F(w)$, N rejects w.

- This shows that N is a decider for A
- Therefore A is decidable

Mapping Reducibility

Theorem 5.22

If $A \leq_m B$ and B is decidable, then A is decidable.

- Let the following:
  - $P \equiv A \leq_m B$
  - $Q \equiv B$ is decidable
  - $R \equiv A$ is decidable
- Since $(P \wedge Q) \rightarrow R \equiv (P \wedge \neg R) \rightarrow \neg Q$, we have

Corollary 5.23

If $A \leq_m B$ and A is undecidable, then B is undecidable.

Mapping Reducibility

Corollary 5.23

If $A \leq_m B$ and A is undecidable, then B is undecidable.

- We already know from the fact that A_{TM} is undecidable
- Because of the above Corollary, to show that a language B is undecidable, we just need to show that $A_{\mathrm{TM}} \leq_m B$
  - A_{TM} can be replaced by other undecidable language such as E_{TM}
  - To show that $A_{\mathrm{TM}} \leq_m B$, we need to construct a TM F that behaves like a reduction from A_{TM} to B satisfying

$$\langle M, w \rangle \in A_{\mathrm{TM}} \leftrightarrow F(\langle M, w \rangle) \in B$$

  - Also need to prove that F actually satisfies the above property

$HALT_{\text{TM}}$ is undecidable

A_{TM} and $HALT_{\text{TM}}$

$$A_{\text{TM}} = \{\langle M, w\rangle \mid M \text{ is a TM and } M \text{ accepts } w\}$$
$$HALT_{\text{TM}} = \{\langle M, w\rangle \mid M \text{ is a TM and } M \text{ halts on input } w\}$$

- To use mapping reducibility to show that $HALT_{\text{TM}}$ is undecidable:
  - We need to show that A_{TM} is mapping reducible to $HALT_{\text{TM}}$ ($A_{\text{TM}} \leq_m HALT_{\text{TM}}$)
  - Since $A_{\text{TM}} \leq_m HALT_{\text{TM}}$ and A_{TM} is undecidable, from Corollary 5.23, $HALT_{\text{TM}}$ is undecidable
- To show that $A_{\text{TM}} \leq_m HALT_{\text{TM}}$, we need a computable function $F : \Sigma^* \to \Sigma^*$ (TM F) such that

$$\langle M, w\rangle \in A_{\text{TM}} \leftrightarrow F(\langle M, w\rangle) \in HALT_{\text{TM}}$$

- Note that $F(\langle M, w\rangle)$ must be in the form of $\langle M', w'\rangle$ (the format of $HALT_{\text{TM}}$)

$HALT_{\text{TM}}$ is undecidable

- A computable function F where $F(\langle M, w\rangle) = \langle M', w'\rangle$ is a TM
  - A TM that takes $\langle M, w\rangle$ as its input on its tape and produces the output $\langle M', w'\rangle$ on its tape after it is halted
- Note that TM F must be a reduction from A_{TM} to $HALT_{\text{TM}}$:

$$\langle M, w\rangle \in A_{\text{TM}} \leftrightarrow F(\langle M, w\rangle) \in HALT_{\text{TM}}$$

  where $F(\langle M, w\rangle) = \langle M', w'\rangle$
  - $\langle M, w\rangle \in A_{\text{TM}} \rightarrow F(\langle M, w\rangle) \in HALT_{\text{TM}}$
    $\rightsquigarrow M$ accepts $w \rightarrow M'$ halts on input w'
  - $\langle M, w\rangle \notin A_{\text{TM}} \rightarrow F(\langle M, w\rangle) \notin HALT_{\text{TM}}$
    $\rightsquigarrow M$ does not accept $w \rightarrow M'$ does not halt on input w'

$HALT_{\text{TM}}$ is Undecidable

- Machine F computes a reduction f:

  $F =$"On input $\langle M, w \rangle$ where M is a TM and w is a string:

  1. Construct the following machine M':

     $M' =$"On input x:

     1. Run M on w.
     2. If M accepts w, *accept*. If M rejects w, enter infinite loop."

  2. Output $\langle M', \varepsilon \rangle$."

- Next, we need to check whether TM F satisfies

$$\langle M, w \rangle \in A_{\text{TM}} \leftrightarrow \langle M', w' \rangle \in HALT_{\text{TM}}$$

  where $\langle M', \varepsilon \rangle = F(\langle M, w \rangle)$ (the output of TM F)

- To do so, we need to show the following:

  1. If $\langle M, w \rangle \in A_{\text{TM}}$, $\langle M', \varepsilon \rangle \in HALT_{\text{TM}}$
  2. If $\langle M, w \rangle \notin A_{\text{TM}}$, $\langle M', \varepsilon \rangle \notin HALT_{\text{TM}}$

$HALT_{\text{TM}}$ is Undecidable

- Show that

$$\langle M, w \rangle \in A_{\text{TM}} \leftrightarrow F(\langle M, w \rangle) \in HALT_{\text{TM}}$$

  where $F(\langle M, w \rangle) = \langle M', \varepsilon \rangle$
  - Assume that $\langle M, w \rangle \in A_{\text{TM}}$. Since $\langle M, w \rangle \in A_{\text{TM}}$, M is a TM that accepts w. Since M accepts w, from the definition of TM M', M' halts (accepts) on all strings including ε. Since M' halts on ε, $\langle M', \varepsilon \rangle \in HALT_{\text{TM}}$. Since $\langle M', \varepsilon \rangle = F(\langle M, w \rangle)$, $F(\langle M, w \rangle) \in HALT_{\text{TM}}$.
  - Assume that $\langle M, w \rangle \notin A_{\text{TM}}$. Since $\langle M, w \rangle \notin A_{\text{TM}}$, M is a TM that does not accept w. Since M does not accept w, from the definition of TM M', M' does not halt on all strings including ε. Since M' does not halt on ε, $\langle M', \varepsilon \rangle \notin HALT_{\text{TM}}$. Since $\langle M', \varepsilon \rangle = F(\langle M, w \rangle)$, $F(\langle M, w \rangle) \notin HALT_{\text{TM}}$.
- This show that TM F is a reduction from A_{TM} to $HALT_{\text{TM}}$
- In other words, $A_{\text{TM}} \leq_m HALT_{\text{TM}}$
- Since A_{TM} is undecidable, $HALT_{\text{TM}}$ is undecidable (Corollary 5.23)

$HALT_{\text{TM}}$ is Undecidable

- Let's compare two methods:
- Method #1: We construct a TM S to decide A_{TM} as follows:

  $S =$ "On input $\langle M, w \rangle$ where M is a TM and w is a string:
  1. Construct a TM M' as follows:

     $M' =$ "On input x:
     1. Run M on input w.
     2. If M accepts w, *accept*. If M rejects w, enter infinite loop."
  2. Run R (a decider for $HALT_{\text{TM}}$) on input $\langle M', \varepsilon \rangle$.
  3. If R accepts $\langle M', \varepsilon \rangle$, *accept*. If R rejects $\langle M', \varepsilon \rangle$, *reject*."
- Method #2: Construct a reduction F from A_{TM} to $HALT_{\text{TM}}$ as follows:

  $F =$ "On input $\langle M, w \rangle$ where M is a TM and w is a string:
  1. Construct a TM M' as follows:

     $M' =$ "On input x:
     1. Run M on input w.
     2. If M accepts w, *accept*. If M rejects w, enter infinite loop."
  2. Output $\langle M', \varepsilon \rangle$."
- Did you see the similarity?

E_{TM} is Undecidable

- To show that E_{TM} is undecidable using the mapping reducibility, we need to show that $A_{\mathrm{TM}} \leq_m E_{\mathrm{TM}}$
- In other word, we need to find a computable function $F : \Sigma^* \to \Sigma^*$ (TM F) such that

$$\langle M, w \rangle \in A_{\mathrm{TM}} \leftrightarrow F(\langle M, w \rangle) \in E_{\mathrm{TM}}$$

- Suppose we construct the following TM F:

  $F =$ "On input $\langle M, w \rangle$ where M is a TM and w is a string:

  1. Construct TM M' as follows:

     $M' =$ "On input x:

     1. Run M on input w.
     2. If M accepts w, *accept*. If M rejects w, *reject*."

  2. Output $\langle M' \rangle$."

- The above TM F does not satisfy the above property
- It satisfies

$$\langle M, w \rangle \in A_{\mathrm{TM}} \leftrightarrow F(\langle M, w \rangle) \in \overline{E_{\mathrm{TM}}}$$

E_{TM} is Undecidable

- TM F is a reduction from A_{TM} to $\overline{E_{\mathrm{TM}}}$
- This shows that $A_{\mathrm{TM}} \leq_m \overline{E_{\mathrm{TM}}}$
- By Corollary 5.23, since A_{TM} is undecidable, $\overline{E_{\mathrm{TM}}}$ is undecidable
- Note that if $\overline{E_{\mathrm{TM}}}$ is decidable, E_{TM} is decidable
  - We discussed this toward the end of Chapter 4
- Therefore, E_{TM} is undecidable

$REGULAR_{\mathrm{TM}}$ is Undecidable

Use the **mapping reducibility** to show that
$REGULAR_{\mathrm{TM}} = \{\langle M \rangle \mid M$ is a TM and $L(M)$ is regular$\}$ is undecidable.

Solution: We need to construct a reduction F from A_{TM} to $REGULAR_{\mathrm{TM}}$ satisfying

$$\langle M, w \rangle \in A_{\mathrm{TM}} \leftrightarrow F(\langle M, w \rangle) \in REGULAR_{\mathrm{TM}}$$

Construct a TM F as follows:

$F =$ "On input $\langle M, w \rangle$ where M is a TM and w is a string:

1. Construct a TM M' as follows:

   $M' =$ "On input x:

   1. If x is in the form of $0^n 1^n$, *accept*.
   2. If x is not in the form of $0^n 1^n$, run M on input w. If M accepts w, *accept*; otherwise, *reject*."
2. Output $\langle M' \rangle$."

Next we need to show that the above TM F satisfies the property:

$$\langle M, w \rangle \in A_{\mathrm{TM}} \leftrightarrow \langle M' \rangle \in REGULAR_{\mathrm{TM}}$$

where $\langle M' \rangle = F(\langle M, w \rangle)$ (output of TM F):

$REGULAR_{\text{TM}}$ is Undecidable

Continue:

- If $\langle M, w \rangle \in A_{\text{TM}}$, $F(\langle M, w \rangle) \in REGULAR_{\text{TM}}$:

  Assume that $\langle M, w \rangle \in A_{\text{TM}}$. Since $\langle M, w \rangle \in A_{\text{TM}}$, M is a TM that accepts w. Since M accepts w, from the definition of TM M', $L(M') = \Sigma^*$. Since $L(M') = \Sigma^*$, $L(M')$ is regular. Since $L(M')$ is regular, $\langle M' \rangle \in REGULAR_{\text{TM}}$. Since $F(\langle M, w \rangle) = \langle M' \rangle$, $F(\langle M, w \rangle) \in REGULAR_{\text{TM}}$.

- If $\langle M, w \rangle \notin A_{\text{TM}}$, $F(\langle M, w \rangle) \notin REGULAR_{\text{TM}}$:

  Assume that $\langle M, w \rangle \notin A_{\text{TM}}$. Since $\langle M, w \rangle \notin A_{\text{TM}}$, M is a TM that does not accept w. Since M does not accept w, from the definition of TM M', $L(M') = \{0^n 1^n \mid n \geq 0\}$. Since $L(M') = \{0^n 1^n \mid n \geq 0\}$, $L(M')$ is not regular. Since $L(M')$ is not regular, $\langle M' \rangle \notin REGULAR_{\text{TM}}$. Since $F(\langle M, w \rangle) = \langle M' \rangle$, $F(\langle M, w \rangle) \notin REGULAR_{\text{TM}}$.

This show that TM F is a reduction from A_{TM} to $REGULAR_{\text{TM}}$. Therefore $A_{\text{TM}} \leq_m REGULAR_{\text{TM}}$. Since A_{TM} is undecidable, by Corollary 5.23, $REGULAR_{\text{TM}}$ is undecidable.

EQ_{TM} is Undecidable

- We need to show that $E_{\mathrm{TM}} \leq_m EQ_{\mathrm{TM}}$
- We need to find a computable function $F : \Sigma^* \to \Sigma^*$ such that

$$\langle M \rangle \in E_{\mathrm{TM}} \leftrightarrow F(\langle M \rangle) \in EQ_{\mathrm{TM}}.$$

  where $F(\langle M \rangle) = \langle M_1, M_2 \rangle$ for some TMs M_1 and M_2
- Here is the TM F:

  $F =$ "On input $\langle M \rangle$ where M is a TM:
  1. Construct a TM M_{E} as follows:

     $M_{\mathrm{E}} =$ "On input x:
     1. *reject*."
  2. Output $\langle M, M_{\mathrm{E}} \rangle$."
- After proofs, we can make a conclusion that F is a reduction from E_{TM} to EQ_{TM}
- Therefore, $E_{\mathrm{TM}} \leq_m EQ_{\mathrm{TM}}$
- Since E_{TM} is undecidable, by Corollary 5.23, EQ_{TM} is undecidable

Undecidability of the Post Correspondence Problem

- Earlier, we construct an instance P' of $MPCP$ that has a match if and only if M accepts w
- Formally,

$$\langle M, w \rangle \in A_{\text{TM}} \leftrightarrow F(\langle M, w \rangle) \in MPCP$$

  where $F(\langle M, w \rangle) = P'$
- This shows that $A_{\text{TM}} \leq_m MPCP$
  - Since A_{TM} is undecidable, $MPCP$ is undecidable
- Then we construct an instance P of PCP that has a match if and only if P' has a matchFormally,

$$P' \in MPCP \leftrightarrow F'(P') \in PCP.$$

  where $F'(P') = P$
- This shows that $MPCP \leq_m PCP$
  - Since $MPCP$ is undecidable, PCP is undecidable

Conclusion

- The **mapping reducibility** can be used to show that a language is undecidable
- Corollary 5.23

  If $A \leq_m B$ and A is undecidable, then B is undecidable
- Since A_{TM} is undecidable, to show that a language B is undecidable, simply show that $A_{\mathrm{TM}} \leq_m B$
- Need to show that a **reduction** F satisfying

$$\langle M, w \rangle \in A_{\mathrm{TM}} \leftrightarrow F(\langle M, w \rangle) \in B$$

  exists
- Instead of A_{TM}, other known undecidable language can be used